COLLABORATIVE DEVELOPMENT PLATFORM

A-SW HUB

Integrated Collaborative Platform for Autonomous Agricultural Machinery Software Development

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PROJECT

Open-Source A-SW and Collaborative Development Service Platform for Digital Transformation of Agricultural Machinery

SUB-PROJECT

Development and Demonstration of Open A-SW for Unmanned Autonomous Agricultural Operations

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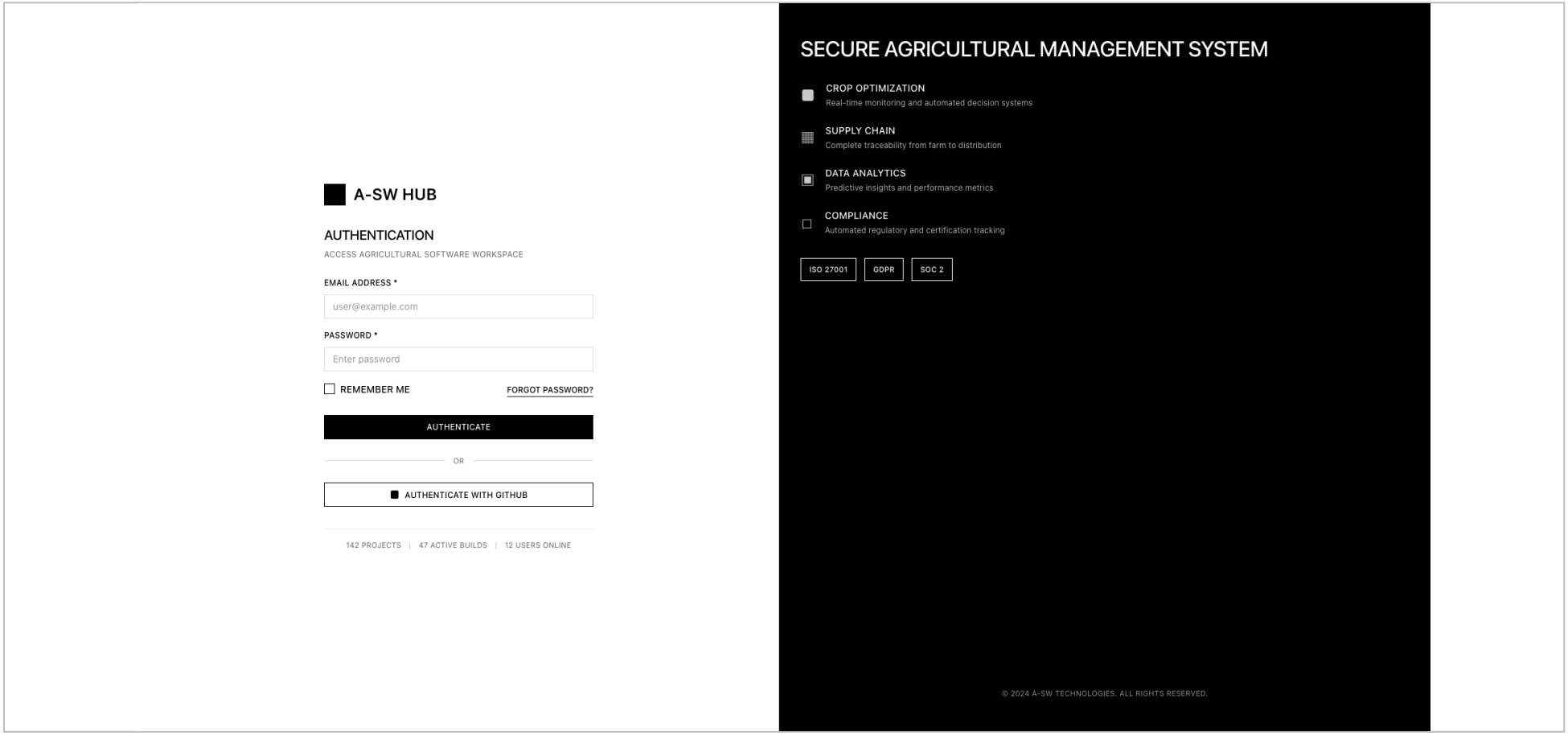
Login

PURPOSE

Secure user authentication gateway for accessing the A-SW Hub platform with role-based credentials and session management.

DESCRIPTION

Split-screen authentication interface with white form area (email/password, GitHub OAuth) on left and black pitch area showcasing platform features on right. Includes session persistence and real-time statistics display.



Login

OVERVIEW

Split-screen authentication interface with white form area (email/password, GitHub OAuth) on left and black pitch area showcasing platform features on right. Includes session persistence and real-time statistics display.

PURPOSE

Secure user authentication gateway for accessing the A-SW Hub platform with role-based credentials and session management.

KEY FEATURES

- Secure credential validation with authStore integration
- Session token generation and management
- Role-based access control (Admin, Developer, Viewer)
- GitHub OAuth authentication support
- Session persistence with localStorage integration
- Real-time authentication status feedback with loading states
- Responsive design with mobile optimization (pitch area hidden on mobile)
- Industrial B/W design system with consistent typography
- Comprehensive form validation and error handling
- Automatic navigation to dashboard on successful login

COMPONENTS

COMPONENT
Email input field with HTML5 validation and autocomplete
Password input field with visibility toggle and autocomplete
Remember me checkbox for session persistence
Primary authentication button with loading state
GitHub OAuth integration button with ASCII icon
Platform statistics footer (projects, builds, online users)
Split-screen layout: white form area (left) + black pitch area (right)
A-SW HUB logo with geometric block icon
Forgot password link with underline styling
Security badges display (ISO 27001, GDPR, SOC 2)
Feature showcase cards (Crop Optimization, Supply Chain, Data Analytics, Compliance)
Real-time error message display with monochrome styling

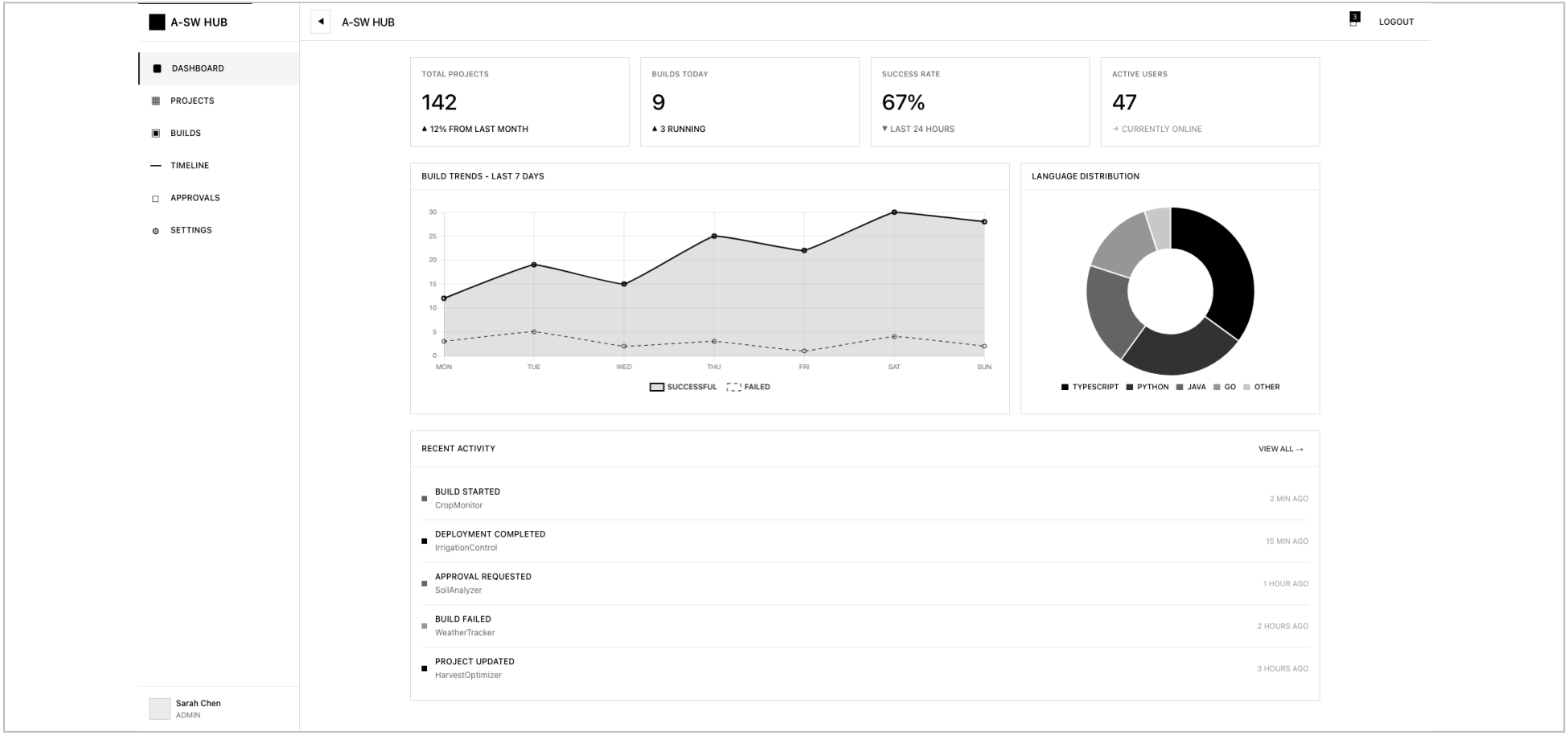
Dashboard

PURPOSE

Centralized overview of all projects, builds, issues, and team activities across the A-SW Hub platform with real-time statistics and interactive visualizations.

DESCRIPTION

Main landing page displaying four statistics cards (projects, builds, success rate, users), two Chart.js visualizations (7-day build trends, language distribution), and real-time activity feed with recent platform events.



Dashboard

OVERVIEW

Main landing page displaying four statistics cards (projects, builds, success rate, users), two Chart.js visualizations (7-day build trends, language distribution), and real-time activity feed with recent platform events.

PURPOSE

Centralized overview of all projects, builds, issues, and team activities across the A-SW Hub platform with real-time statistics and interactive visualizations.

KEY FEATURES

- Real-time data updates from buildsStore and projectsStore
- Role-based content filtering based on user permissions
- Interactive Chart.js visualizations with monochrome styling
- Build metrics calculation (total, running, failed, success rate)
- Activity feed with status-based color coding and icons
- Automatic chart cleanup on component destruction
- Dynamic statistics based on actual build data
- Responsive design with grid collapsing on smaller screens
- Hover interactions on charts for detailed data points
- Background pulse animation for running builds in activity feed

COMPONENTS

COMPONENT
Four statistics cards with trend indicators (up/down/neutral arrows)
Line chart displaying build trends (successful vs failed) over 7 days using Chart.js
Doughnut chart showing language distribution across projects
Real-time activity feed with 5 recent activities
Activity status indicators (running: pulsing animation, success: solid, pending: hollow, failed: subtle)
Panel component wrapper with title and optional actions slot
View All button for accessing complete activity history
Stat component with label, value, trend direction, and trend description
Monochrome chart styling (black lines, fills, and labels)
Responsive grid layout (charts stack on mobile <1024px)

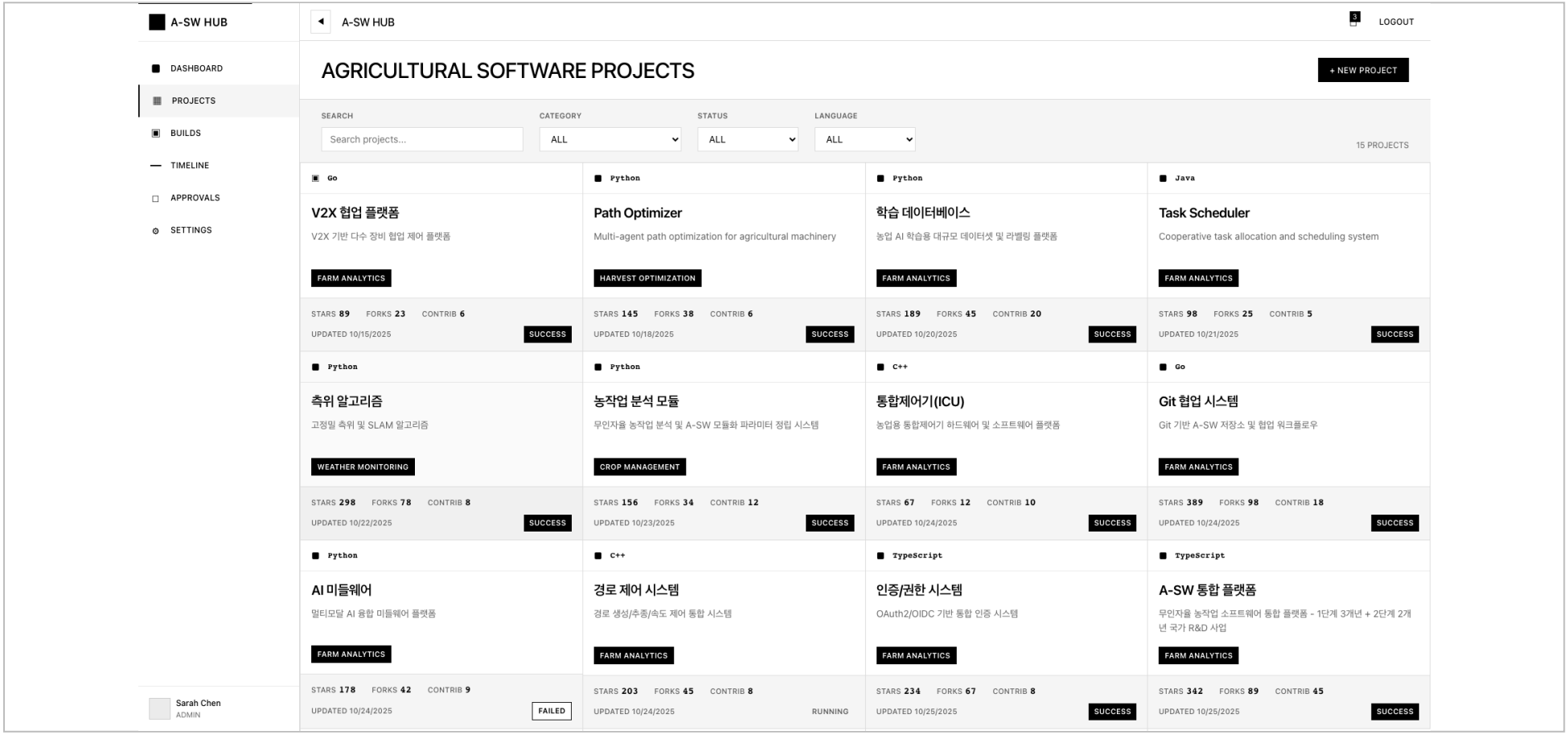
Projects List

PURPOSE

Comprehensive directory of all agricultural software projects with advanced filtering, search capabilities, and grid-based visualization for easy project discovery and navigation.

DESCRIPTION

Grid layout displaying all projects with comprehensive filters (search, category, status, language). Each card shows status with ASCII indicators, language, name, description, category badge, and statistics (stars, forks, contributors).



Projects List

OVERVIEW

Grid layout displaying all projects with comprehensive filters (search, category, status, language). Each card shows status with ASCII indicators, language, name, description, category badge, and statistics (stars, forks, contributors).

PURPOSE

Comprehensive directory of all agricultural software projects with advanced filtering, search capabilities, and grid-based visualization for easy project discovery and navigation.

KEY FEATURES

- Real-time project status updates via projectsStore
- Multi-criteria filtering and search with reactive updates
- Search query binding with input event handling
- Category, status, and language filter combinations
- Filtered projects count from filteredProjects store
- Click navigation to project detail page (/projects/{slug})
- ASCII status indicators based on project state
- Responsive grid with border-based layout (no gaps)
- Hover state with subtle background color change
- Project card height fixed at 280px for consistency
- Build status conditional rendering and styling
- Date formatting for last commit information
- *...and 3 more features*

COMPONENTS

COMPONENT
Page header with title and 'NEW PROJECT' button
Filters bar with search input (300px width with auto-complete)
Category dropdown filter with 12 options (all + 11 project categories)
Status dropdown filter (all, active, maintenance, archived, deprecated)
Programming language dropdown filter (all + 10 languages)
Filter results counter showing number of projects
Responsive projects grid (auto-fill, minmax(400px, 1fr))
Project card with header, body, and footer sections
Status indicator with ASCII symbols based on project state
Language badge in monospace font (JetBrains Mono/Fira Code)
Project name heading with semibold weight
Description text with 2-line ellipsis overflow
Category badge with inverted colors (white text on black background)
Statistics row showing stars, forks, and contributors count
Last commit date with formatted display
Build status badge (success or failed) with conditional styling
Empty state message when no projects match filters

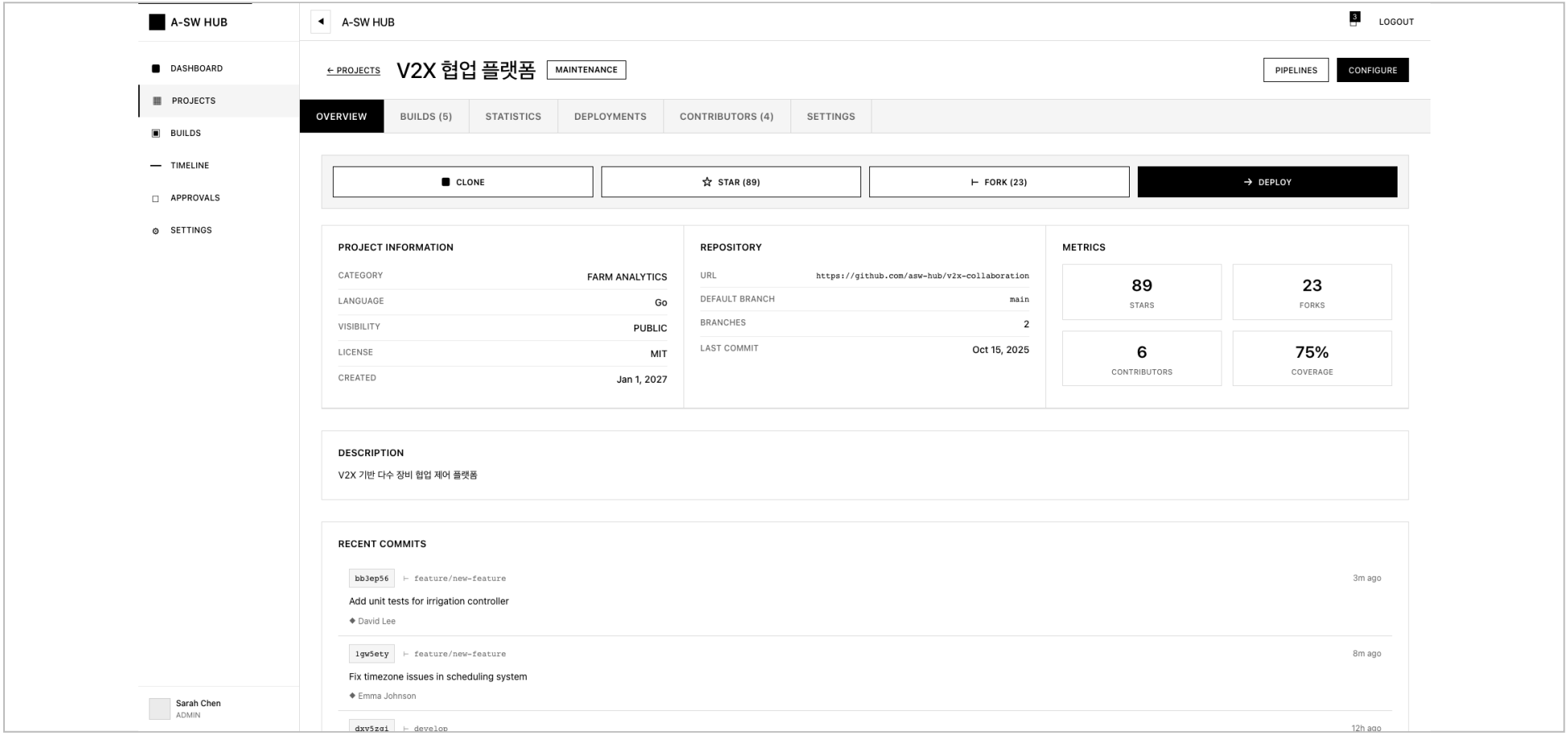
Project Detail

PURPOSE

Comprehensive project management hub with tabbed interface providing complete visibility into project overview, builds, deployments, team members, and configuration settings.

DESCRIPTION

Central project hub with header displaying name, status, and metrics, followed by six-tab navigation (Overview, Builds, Statistics, Deployments, Contributors, Settings) with role-based access control and real-time data updates.



Project Detail

OVERVIEW

Central project hub with header displaying name, status, and metrics, followed by six-tab navigation (Overview, Builds, Statistics, Deployments, Contributors, Settings) with role-based access control and real-time data updates.

PURPOSE

Comprehensive project management hub with tabbed interface providing complete visibility into project overview, builds, deployments, team members, and configuration settings.

KEY FEATURES

- Six specialized tabs for different project aspects
- Real-time data synchronization via Svelte stores
- Role-based tab visibility and access control
- Tab navigation with active state management
- Responsive design with mobile tab stacking
- Keyboard navigation support
- Deep linking to specific tabs via URL hash

COMPONENTS

COMPONENT
Project header with name, status badge (■/▣/□), and action buttons
Six-tab navigation bar with active state indicators
Tab content area with conditional rendering based on activeTab
Project metrics badges (stars, forks, issues, commits)
Real-time status indicators
Action button group in header (Edit, Settings, Pipeline)
Responsive layout with mobile optimization

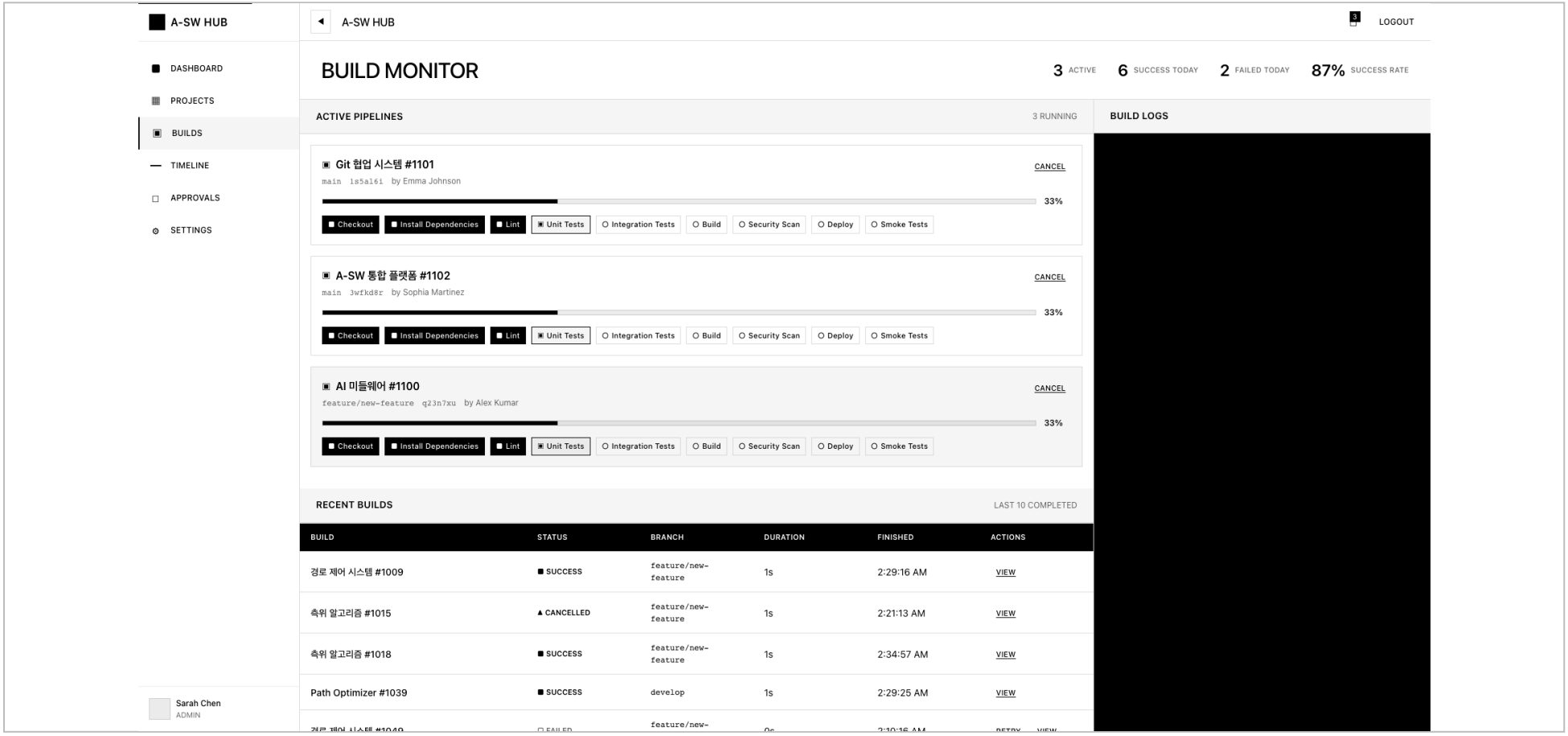
Builds Dashboard

PURPOSE

Real-time monitoring of all CI/CD build pipelines across projects with comprehensive status tracking, log visualization, and build management capabilities.

DESCRIPTION

Split-view interface with active builds panel (real-time progress, 6-stage status) and terminal-style log viewer (collapsible sections, color-coded). Header shows statistics, bottom table displays 10 recent builds with actions.



Builds Dashboard

OVERVIEW

Split-view interface with active builds panel (real-time progress, 6-stage status) and terminal-style log viewer (collapsible sections, color-coded). Header shows statistics, bottom table displays 10 recent builds with actions.

PURPOSE

Real-time monitoring of all CI/CD build pipelines across projects with comprehensive status tracking, log visualization, and build management capabilities.

KEY FEATURES

- Multi-project build monitoring with centralized view
- Real-time build status tracking via WebSocket updates
- Build progress calculation across all stages
- Stage-by-stage execution visualization
- Terminal-style log viewer with monospace font
- Collapsible log sections for organized viewing
- Syntax-highlighted logs with color coding by level
- Build cancellation for running pipelines
- Failed build retry functionality
- Build history with last 10 completions
- Sortable table columns for easy navigation
- Duration formatting (hours, minutes, seconds)
- *...and 8 more features*

COMPONENTS

COMPONENT
Page header with 'BUILD MONITOR' title and statistics (4 stat cards)
Statistics: Active count, Success Today, Failed Today, Success Rate percentage
Split-view layout: Active builds section (left, scrollable) + Log viewer (right, fixed 500px)
Active pipelines section with pipeline cards (dashed border animation for in_progress)
Pipeline card: project name, build number, branch, commit SHA (7 chars), triggered by user
Progress bar with percentage and animated fill transition
Stage indicators: 6 stages with status icons (✓ passed, ✕ failed, ◌ running, ○ pending)
Stage status badges with appropriate colors and icons
Cancel button for running builds (stops execution)
Recent builds table with 6 columns (Build, Status, Branch, Duration, Finished, Actions)
Table header with inverted colors (white on black)
Status badges in table rows with ASCII icons
Retry button for failed builds (visible only on failure)
View button navigating to detailed build page
Log viewer with black terminal background (#000000)
Collapsible log sections with expand/collapse controls
Expand All / Collapse All buttons in log header
Log sections with status-specific border colors
Log lines with syntax highlighting (ERROR: red, WARNING: yellow, SUCCESS: green, INFO: cyan)
Section headers with toggle icon (▼ expanded, ▶ collapsed), status icon, name, duration

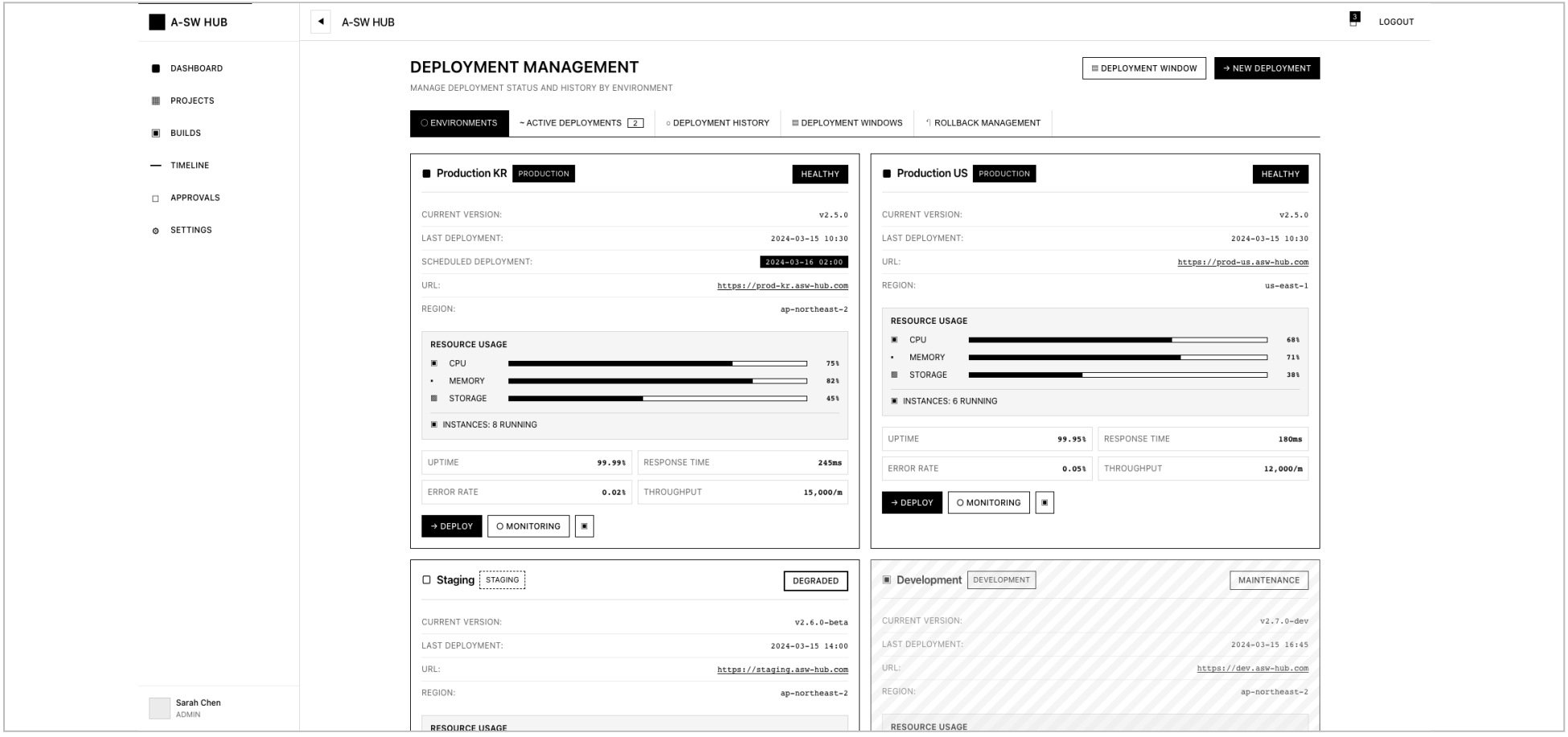
Deployments

PURPOSE

Comprehensive deployment management dashboard showing deployment status, history, and controls across all environments (development, staging, production) with rollback capabilities and deployment window scheduling.

DESCRIPTION

5-tab interface showing: (1) Environment cards with status, metrics, and resource usage; (2) Active deployments with 6-stage progress; (3) Filterable deployment history; (4) Deployment windows scheduling; (5) Rollback management for last 30 days.



Deployments

OVERVIEW

5-tab interface showing: (1) Environment cards with status, metrics, and resource usage; (2) Active deployments with 6-stage progress; (3) Filterable deployment history; (4) Deployment windows scheduling; (5) Rollback management for last 30 days.

PURPOSE

Comprehensive deployment management dashboard showing deployment status, history, and controls across all environments (development, staging, production) with rollback capabilities and deployment window scheduling.

KEY FEATURES

- Multi-environment deployment tracking (4 environments)
- Real-time deployment status monitoring
- Resource usage visualization with progress bars
- Performance metrics display (uptime, response time, error rate, throughput)
- One-click deployment triggers per environment
- Six-stage deployment pipeline with progress tracking
- Deployment pause and resume functionality
- Automated rollback capabilities for failed deployments
- Deployment history with filtering (environment, project, status)
- Export deployment history to CSV/JSON
- Deployment window scheduling for planned releases
- Environment-specific deployment settings
- *...and 12 more features*

COMPONENTS

COMPONENT
Page header with title 'DEPLOYMENT MANAGEMENT' and action buttons (Deployment Window, New Deployment)
Five-tab navigation (Environments, Active Deployments, History, Windows, Rollback)
Tab badges showing active deployment count
Environment cards grid (2 columns) with status, version, and metrics
Environment status indicator (HEALTHY: filled, DEGRADED: bordered, DOWN: strikethrough, MAINTENANCE: pattern)
Environment type badges (PRODUCTION: filled black, STAGING: dashed border, DEVELOPMENT: gray)
Resource usage section with 3 progress bars (CPU, Memory, Storage) and percentage labels
Instances count display with running status
Environment metrics grid (2x2): Uptime %, Response Time ms, Error Rate %, Throughput/min
Environment action buttons (Deploy, Monitoring, Menu)
Active deployment cards with real-time status
Deployment status badge with spinner animation for in_progress
6-stage deployment checks with status icons (✓ passed, ✕ failed, ◌ running, ○ pending)
Deployment metadata (deployer, approver, start time, elapsed time)
Expandable deployment details with changelog and artifacts
Deployment action buttons (Pause/Resume, Rollback, Logs)
History filters (Environment, Project, Status dropdowns + Export button)

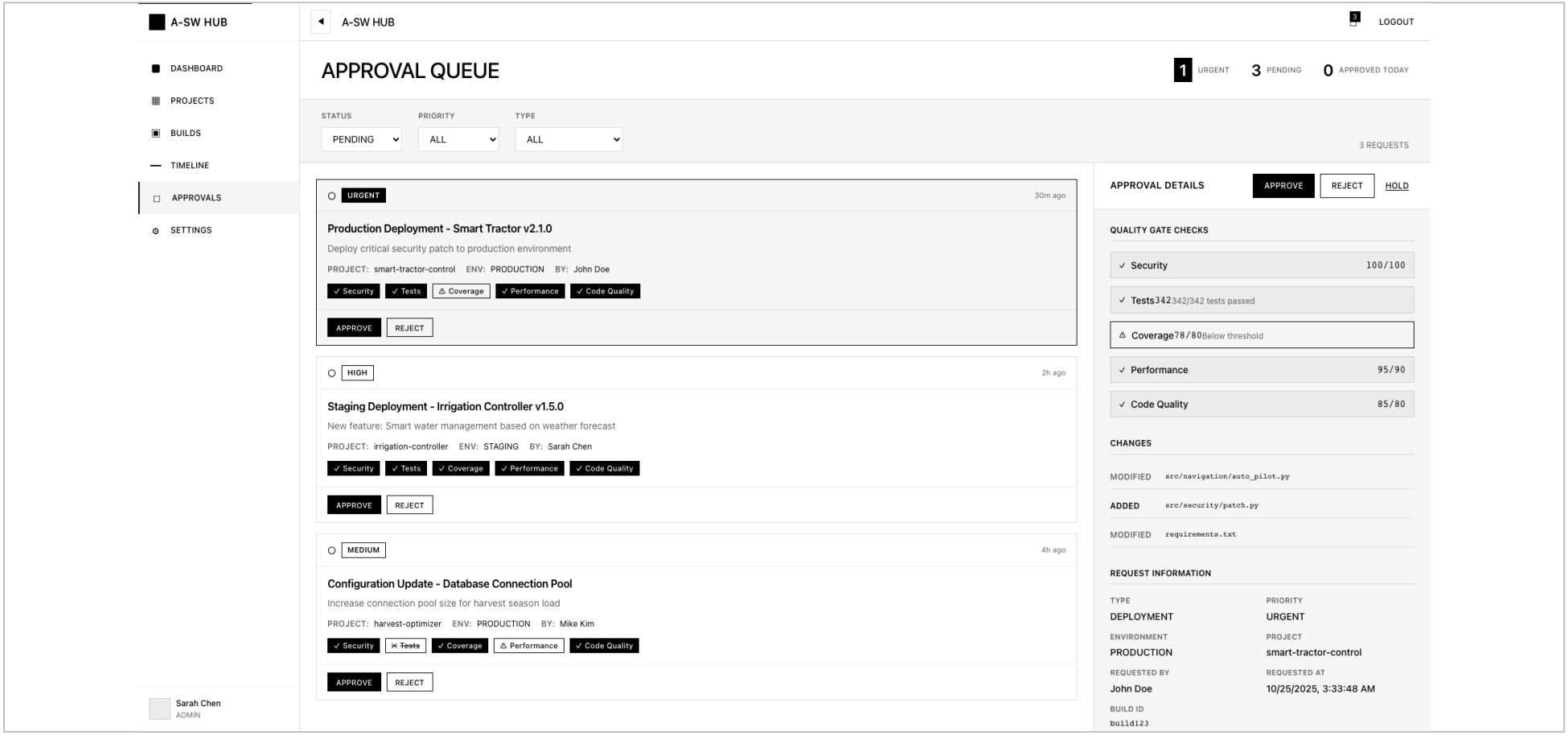
Approvals

PURPOSE

Centralized approval workflow management for deployment requests, merge requests, and configuration changes with comprehensive quality gate checks and priority-based queue management.

DESCRIPTION

Split-view approval management with queue cards (priority, type, quality checks) on left and detailed panel on right (quality gates, file changes, metadata, actions). Header shows statistics, filters enable sorting.



Approvals

OVERVIEW

Split-view approval management with queue cards (priority, type, quality checks) on left and detailed panel on right (quality gates, file changes, metadata, actions). Header shows statistics, filters enable sorting.

PURPOSE

Centralized approval workflow management for deployment requests, merge requests, and configuration changes with comprehensive quality gate checks and priority-based queue management.

KEY FEATURES

- Centralized approval management across all projects
- Priority-based queue (URGENT, HIGH, MEDIUM, LOW)
- Five approval types (deployment, merge, release, configuration, access)
- Comprehensive quality gate checks with pass/warn/fail states
- Quality metrics display (Security, Tests, Coverage, Performance, Code Quality)
- Threshold-based quality validation
- File changes visualization with type indicators
- Approval action workflow (Approve, Reject with reason, Hold)
- Rejection reason requirement with textarea dialog
- Real-time approval status updates
- Time ago formatting (minutes, hours, days)
- Multi-criteria filtering (status, priority, type)
- *...and 11 more features*

COMPONENTS

COMPONENT
Page header with title 'APPROVAL QUEUE' and 3 statistics cards (Urgent: highlighted, Pending, Approved Today)
Filters bar with 3 dropdowns (Status, Priority, Type) and results counter
Split-view layout: Approval list (left, scrollable) + Approval details (right, 500px fixed)
Approval cards in queue with selectable/selected states
Card header: status icon (○ pending, ■ approved, □ rejected, ▣ on-hold), priority badge, time ago
Priority badges with styling (URGENT: black background, HIGH: bordered, MEDIUM/LOW: normal)
Card body: title (16px semibold), description (14px muted), metadata row (project, env, requestor)
Quality checks row: 5 check badges with status icons and names
Quality check badges (PASS: filled, WARN: bordered, FAIL: strikethrough)
Card actions (visible only for pending): Approve and Reject buttons
Details panel header with 'APPROVAL DETAILS' title and action buttons
Quality gate checks section with expandable rows showing name, value, threshold, details
Quality row styling (PASS: surface background, WARN: bordered, FAIL: fail background)
Changes section listing modified/added/deleted files with syntax highlighting
Change type labels (ADDED: foreground color, MODIFIED: muted, DELETED: strikethrough)
File paths in monospace font
Request information grid (2 columns) showing Type, Priority, Environment, Project, Requested By, Requested At, Build ID
Approve button (primary, green when enabled)

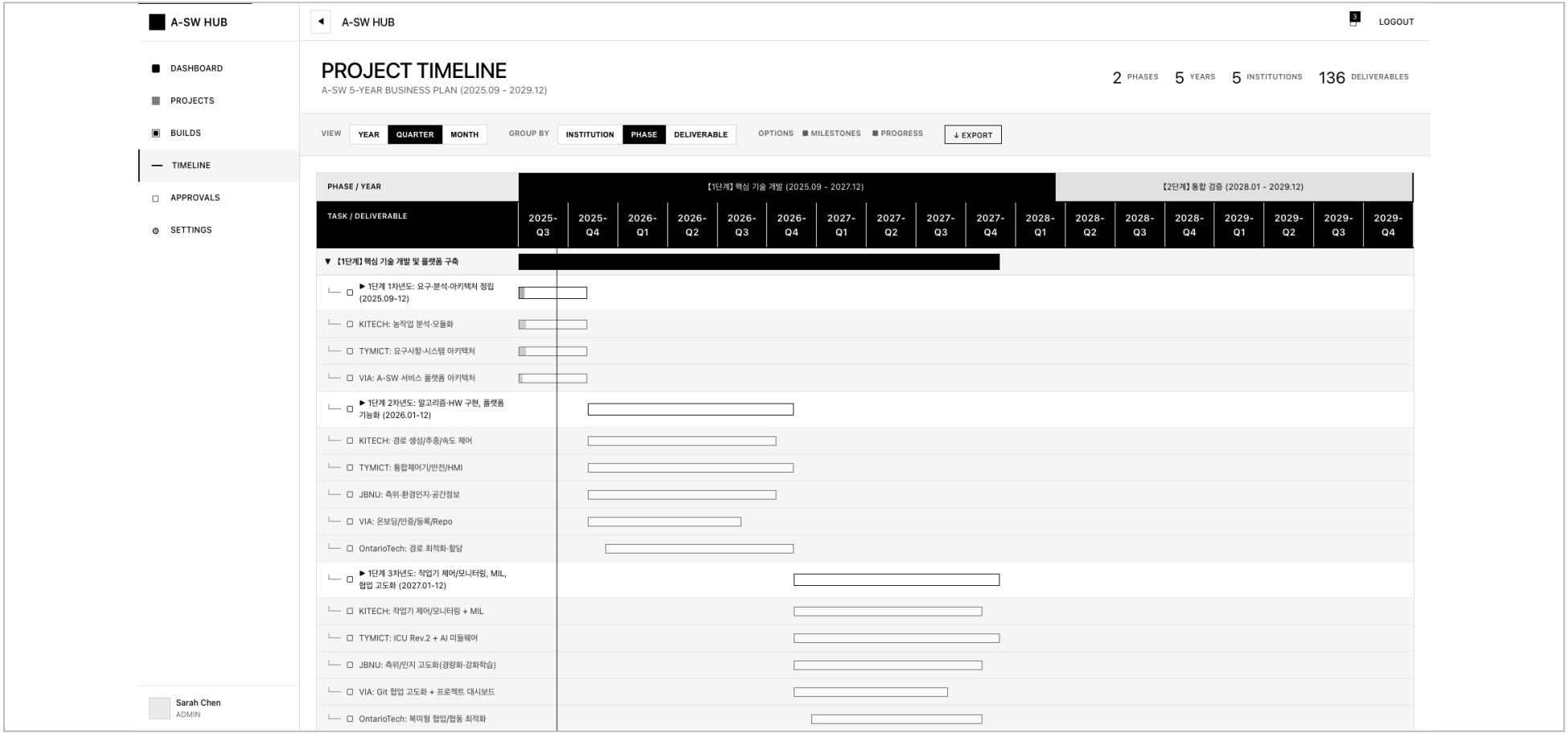
Timeline

PURPOSE

Interactive Gantt chart visualization of project timelines, milestones, and deliverables across the 5-year A-SW initiative (2025.09 - 2029.12) with comprehensive tracking of 136 deliverables for all participating institutions.

DESCRIPTION

Gantt chart for 5-year A-SW plan (2025.09-2029.12) with 136 deliverables across 5 institutions. Features phase indicator, three view modes (YEAR/QUARTER/MONTH), hierarchical tasks with progress bars, TODAY marker, and institution summaries.



Timeline

OVERVIEW

Gantt chart for 5-year A-SW plan (2025.09-2029.12) with 136 deliverables across 5 institutions. Features phase indicator, three view modes (YEAR/QUARTER/MONTH), hierarchical tasks with progress bars, TODAY marker, and institution summaries.

PURPOSE

Interactive Gantt chart visualization of project timelines, milestones, and deliverables across the 5-year A-SW initiative (2025.09 - 2029.12) with comprehensive tracking of 136 deliverables for all participating institutions.

KEY FEATURES

- Five-year timeline visualization (2025.09 - 2029.12)
- 136 deliverables tracking across phases
- Three view modes: YEAR (5 periods), QUARTER (18 quarters), MONTH (52 months)
- Three grouping options: INSTITUTION, PHASE, DELIVERABLE
- Hierarchical task structure with expand/collapse functionality
- Real-time progress tracking with visual indicators
- Milestone visualization with date positioning
- Task dependency relationship visualization
- Gantt data integration from businessPlan module
- Institution filtering (KITECH, TYMICT, JBNU, VIA, OntarioTech)
- Task hierarchy tree view with indentation
- Phase-based color coding (Phase 1: black, Phase 2: gray)
- *...and 11 more features*

COMPONENTS

COMPONENT
Page header with title 'PROJECT TIMELINE' and subtitle showing date range
Statistics panel: 4 stat cards (2 Phases, 5 Years, 5 Institutions, 136 Deliverables)
Controls bar with 3 sections: View mode (YEAR/QUARTER/MONTH buttons), Group by (INSTITUTION/PHASE/DELIVERABLE buttons), Options (Milestones and Progress checkboxes)
Export button for downloading timeline data
Phase indicator bar with 2-segment display (Phase 1: black background, Phase 2: gray background)
Timeline scale header with sticky positioning showing time units based on view mode
Task column header (300px fixed width) with task names and hierarchy
Timeline bars area with horizontal task bars and progress fills
Expandable/collapsible task groups with toggle icons (▼ expanded, ► collapsed)
Group header rows with gray background and border bottom
Child task rows with indent indicators (└──) and status icons
Status icons: ■ completed, ▣ in-progress, □ planned, ▲ delayed
Task bars with different styles: solid (completed), striped (in-progress), hollow (planned), dashed (delayed)
Progress fill overlays (30% opacity) when showProgress is enabled
Milestone diamonds (◆) positioned at specific dates
TODAY marker with vertical line and label
Legend at bottom with 6 items explaining visual indicators
Progress indicator characters: ▓▒▒▒·

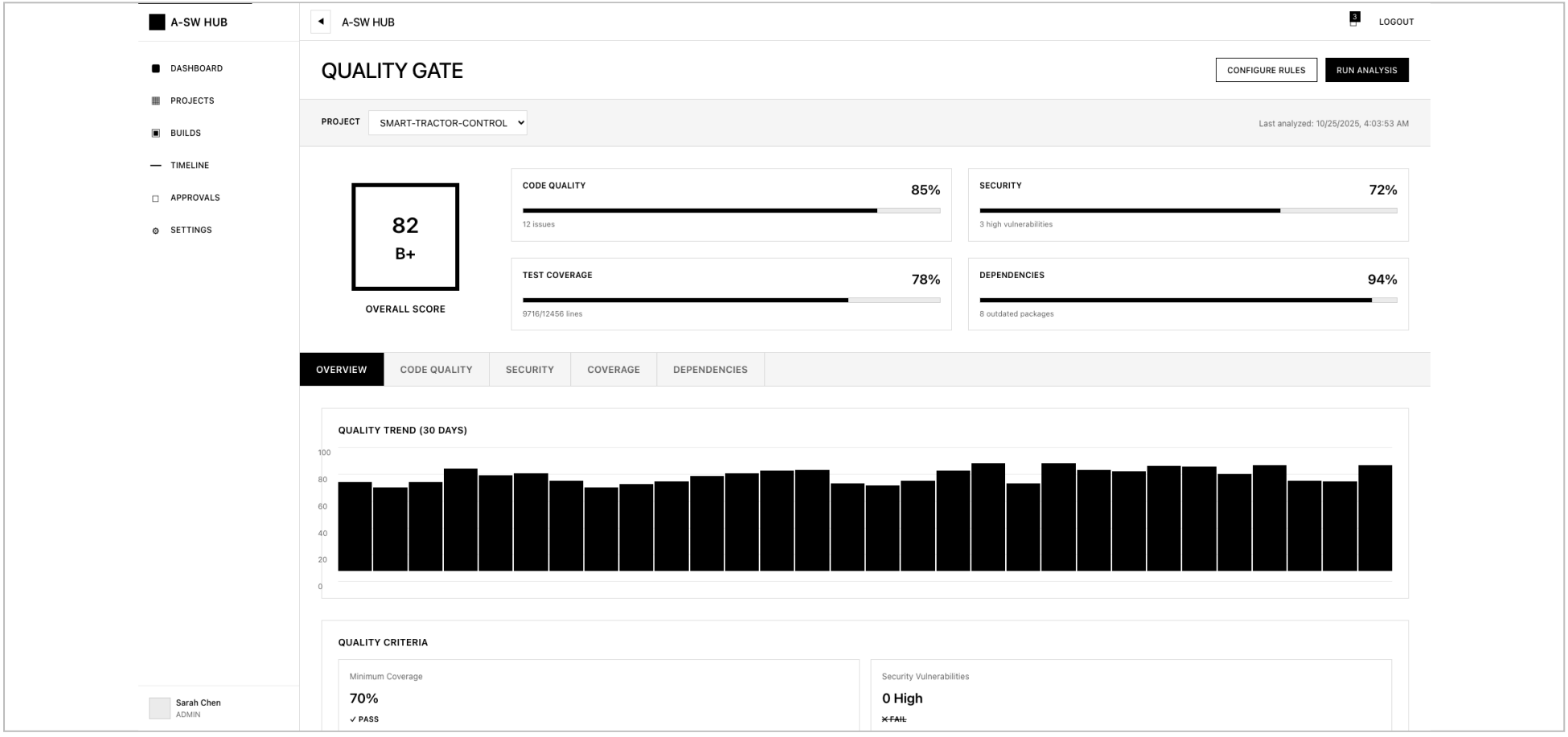
Quality Metrics

PURPOSE

Comprehensive code quality analysis and reporting dashboard providing detailed insights into code quality, security vulnerabilities, test coverage, and dependency health with AI-powered recommendations.

DESCRIPTION

Quality dashboard with overall score (circular indicator, grade) and 4 category scores. Five tabs: (1) 30-day trend, criteria, AI recommendations; (2) Code issues and complexity; (3) Security vulnerabilities; (4) Test coverage; (5) Dependencies and SBOM.



Quality Metrics

OVERVIEW

Quality dashboard with overall score (circular indicator, grade) and 4 category scores. Five tabs: (1) 30-day trend, criteria, AI recommendations; (2) Code issues and complexity; (3) Security vulnerabilities; (4) Test coverage; (5) Dependencies and SBOM.

PURPOSE

Comprehensive code quality analysis and reporting dashboard providing detailed insights into code quality, security vulnerabilities, test coverage, and dependency health with AI-powered recommendations.

KEY FEATURES

- Real-time quality scoring and grading (A-F scale)
- Project-specific quality analysis with historical data
- 30-day quality trend visualization with bar chart
- Code quality metrics (issues, duplication, complexity)
- High complexity function identification with thresholds
- Security vulnerability scanning (high/medium/low classification)
- Security hotspot detection with severity levels
- SQL injection, path traversal, weak crypto detection
- Line and branch coverage measurement
- Uncovered file identification with coverage percentages
- Dependency health monitoring (total, outdated, vulnerable)
- License compatibility analysis
- *...and 11 more features*

COMPONENTS

COMPONENT
Page header with title 'QUALITY GATE' and action buttons (Configure Rules, Run Analysis)
Project selector dropdown with last analysis timestamp
Score dashboard grid: Overall score card (160px circle) + 4 category cards
Overall score circle with large score number (32px), grade letter (24px), border thickness indicating grade
Grade styling (A: filled, B: 6px border, C: 4px border, D: 2px border, F: 2px border)
Category score cards (2x2 grid) with name, percentage, progress bar, metadata
Five-tab navigation (Overview, Code Quality, Security, Coverage, Dependencies)
Overview tab: 30-day trend chart with bars, quality criteria grid (2x2), AI recommendations (3 cards)
Trend chart: 6 horizontal grid lines with labels (0-100), 30 vertical bars with hover tooltips
Quality criteria cards showing criterion name, target value, pass/warn/fail status
AI recommendation cards with emoji icons (💡 refactoring, 🔒 security, 🧪 testing), title, description
Code Quality tab: 3 metric cards (Issues, Duplicate Lines, Complex Functions), high complexity functions table
Functions table: 5 columns (Function, Complexity, File, Line, Action), complexity badges (normal/high: >20)
Security tab: 3 vulnerability count cards (High: filled black, Medium: 2px border, Low: normal), security hotspots list
Hotspot cards with type, severity badge, file location (monospace), description, action buttons (View Code, Fix Suggestion)
Coverage tab: 2 coverage metrics with SVG circular progress rings (120px), metric values inside rings, detailed stats below

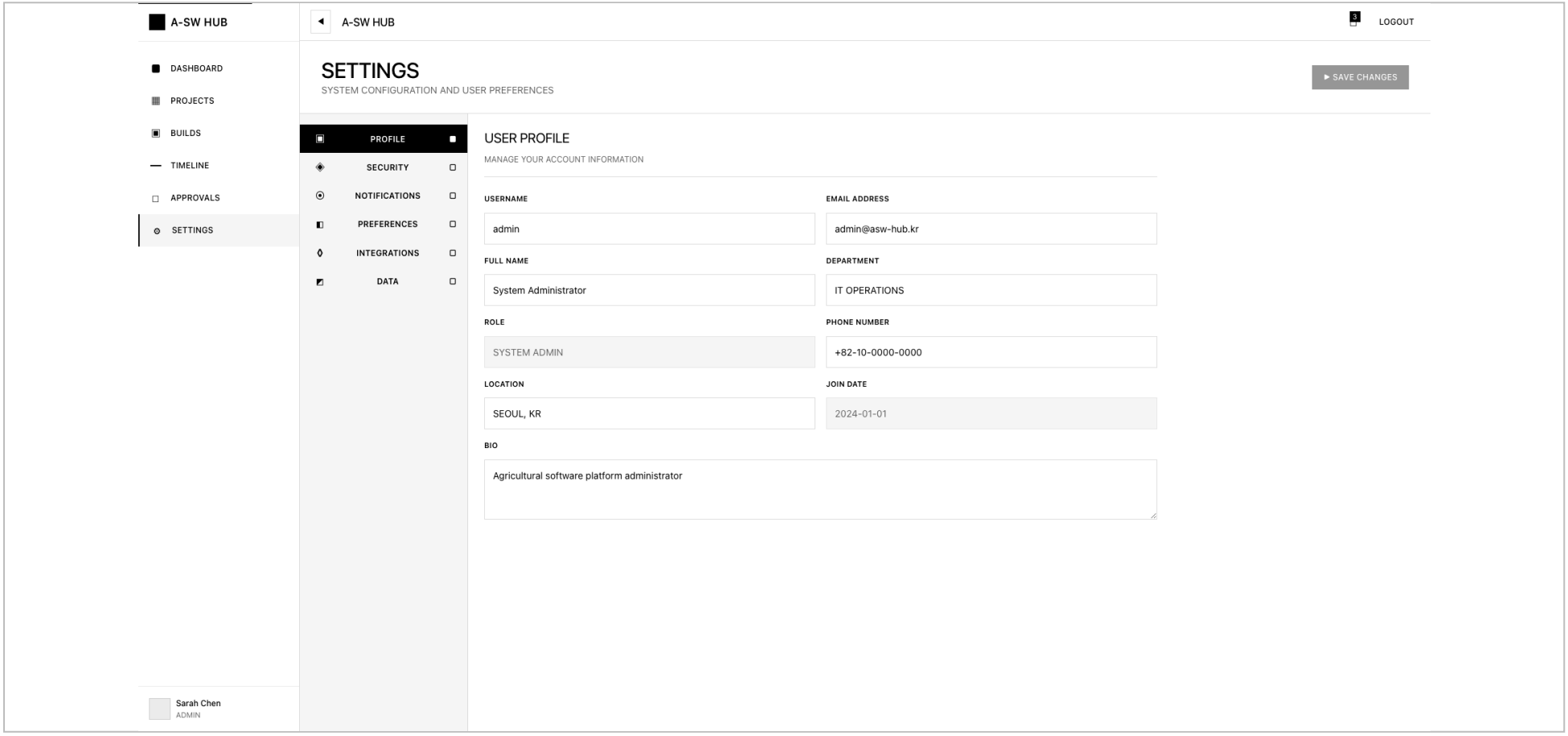
System Settings

PURPOSE

Comprehensive platform-wide settings management for user profile, security, notifications, preferences, integrations, and data privacy with granular control over all system configurations.

DESCRIPTION

Six-category settings with sidebar navigation: Profile, Security, Notifications, Preferences, Integrations, and Data Privacy. Each section has detailed forms for comprehensive platform configuration.



System Settings

OVERVIEW

Six-category settings with sidebar navigation: Profile, Security, Notifications, Preferences, Integrations, and Data Privacy. Each section has detailed forms for comprehensive platform configuration.

PURPOSE

Comprehensive platform-wide settings management for user profile, security, notifications, preferences, integrations, and data privacy with granular control over all system configurations.

KEY FEATURES

- Six-category settings management
- Sidebar navigation with active state
- Form validation with real-time feedback
- Dirty state tracking
- Unsaved changes warning
- Responsive design with collapsible sidebar on mobile

COMPONENTS

COMPONENT
Page header with title 'SETTINGS', subtitle, unsaved changes indicator (● UNSAVED CHANGES), Save Changes button
Sidebar navigation (250px fixed) with 6 tab buttons (Profile, Security, Notifications, Preferences, Integrations, Data)
Tab buttons with ASCII icons (▣ profile, ◈ security, ⦿ notifications, ◧ preferences, ◊ integrations, ◩ data)
Active tab indicator (filled black background, left border)
Tab content area with section-specific forms
Form dirty state tracking indicator
Confirmation dialogs for destructive actions